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Needfinding Study

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BACKGROUND OF THE STUDY

Food waste is a globally acknowledged issue with serious environmental, financial, and food security implications. Households around the world are significantly contributing to this problem, often unwittingly, through their food consumption practices, shopping patterns, food storage, and overall management strategies. The scale of food waste is particularly alarming, representing a crisis that transcends individual household concerns and impacts global food security, economic efficiency, and environmental sustainability. One landmark study by the United Nations' Food and Agriculture Organization in 2013, titled "Food Wastage Footprint: Impacts on Natural Resources" highlighted the sheer scale of the issue. It pointed out that approximately one-third of all food produced for human consumption, which amounts to about 1.3 billion tons per year, is either lost or wasted. These staggering statistics unveil a facet of inefficiency in global food systems, suggesting not just a humanitarian crisis but also a major economic discrepancy. Furthering the economic perspective, the wastage of food represents a substantial loss of resources in terms of manpower, agricultural inputs, transportation, and infrastructural costs. According to the World Resources Institute, the economic value of food lost or wasted annually is approximately $940 billion, a figure that underscores the inefficiency.

The environmental implications add an additional layer of complexity to this issue. When food waste is disposed of in landfills, it undergoes anaerobic decomposition and contributes to the increasing concentration of greenhouse gases in the atmosphere. A 2023 study conducted by the Philippine Department of Science and Technology's Food and Nutrition Research Institute (DOST-FNRI) revealed significant levels of food waste in the country, particularly in urban

areas. This waste not only exacerbates the issue of municipal solid waste management, which is already a challenge in many cities, but also contributes to unnecessary carbon emissions. Within the domestic sphere, factors contributing to food waste are multi-layered and often interconnected. Studies like that of Ganglbauer et al., which focus on the intricacies of domestic food practices, reveal that consumer behavior plays a much bigger role than expected. Factors include poor planning of shopping trips, impulse purchases, incorrect food storage, confusion over labels (best-before and use-by dates), and inefficient or inconsistent food preparation practices. The study also underscores that a lack of understanding or appreciation of food resources can lead to a higher propensity for food wastage at the household level.

Our study seeks to build on these foundational understandings by examining more niche aspects of food management within households. It will explore the dynamics that influence consumer decisions, the efficacy of food storage methods, and the different variables that might affect food utilization. By identifying these key factors and patterns, the study aims to contribute to a more strategic and long-term approach to combating food waste, starting at the household level.

RESEARCH QUESTIONS

This study aims to find out the following:

1. How does a household store and organize their food items?
    a. H1.1: Households store and organize their food items according to its food category (Canned items, dairy, fresh meat, produce, etc.).
    b. H1.2: Products with earlier expiration dates are placed in more accessible areas.
    c. H1.3: Households allocate a specific storage area for each of their members accordingly.
2. How does a household prioritize the consumption of their food items?
    a. H2.1: Households prioritize their consumption based on the expiration dates of items.
    b. H2.2: Households prioritize their consumption of food items based on their cravings.
    c. H2.3: Households prioritize their consumption of food items based on celebratory events that require certain goods.
3. How do specific household characteristics influence food consumption behaviors?
    a. H3.1: The number of household members has a positive relationship with the number of "plates" of food they usually consume.
    b. H3.2: Households with children are more prone to snacking
4. What characteristics of a food item does a household look for when buying groceries?
    a. H4.1: Households generally look for versatile food items that can be utilized in multiple ways.

b. H4.2: Households look for food that satisfies their cravings.

c. H4.3: Households generally look for affordable, high-quantity food items.

d. H4.4: Households generally look for branded, high-quality food items.

e. H4.5: Households incorporate purchasing decisions based on celebratory events, acquiring certain goods for these occasions.

5. What practices contribute to food waste in households?

    a. H5.1: Throwing away spoiled food

    b. H5.2: Throwing away leftover food due to not liking it

    c. H5.3: Disorganization in food storage areas leads to an increase in food waste due to human error.

STUDY PARTICIPANTS

The participants for this study were mainly chosen for their characteristic of being in a household that buys groceries. Survey participants vary in age, monthly household income and their type of home. Interview participants also follow the same standards and in addition to this, they are all from different households. All participants were of legal age.

METHODOLOGY

With regards to understanding problems with food wastes, the researchers focused on understanding household planning and consumption of grocery items. The researchers hypothesized that it was in the household member's actions that contribute greatly to food waste. Therefore, the researchers aimed to find details regarding how they select, store, prioritize, and organize their food items. Household behavior, characteristics, and practices were also taken into account to see how they influence food management.

The data collection method was conducted through an online interview and an online survey. The survey had 31 respondents, and it focused on the quantitative details of household food consumption. A follow-up survey focused on waste management was then sent to the same respondents with 19 responding back. On the other hand, the interview was conducted with 10 participants where they were asked about details regarding food planning, consumption, and management. Statistical analysis was done on the survey results, and thematic analysis was done for the interview transcripts. The researchers aimed to uncover quantitative and qualitative details of the problem as it allows for a better contextual understanding of the survey statistical analysis.

ANALYSIS

Household's Organization and Storage of Food Items

Survey Data Analysis

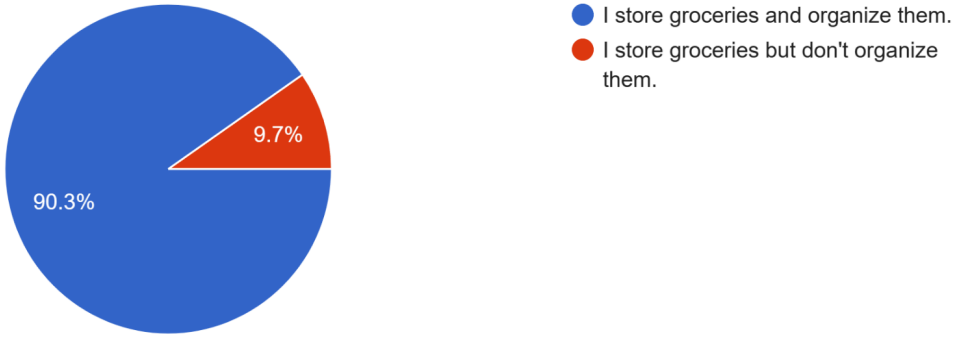


Figure 1.1 *Survey responses when asked "How do you store and organize your groceries?"*

According to the responses from the survey, 92.7% of the respondents (as per Figure 1.1) store and organize their groceries, while 7.3% of the respondents say they store groceries without organizing them. This means that we can presume that it is considered standard behavior for grocery shoppers to organize their groceries when storing them, but there is still an interesting portion among the respondents that say they do not organize their groceries. The reason for this cannot be assumed nor inferred as this requires further investigation.

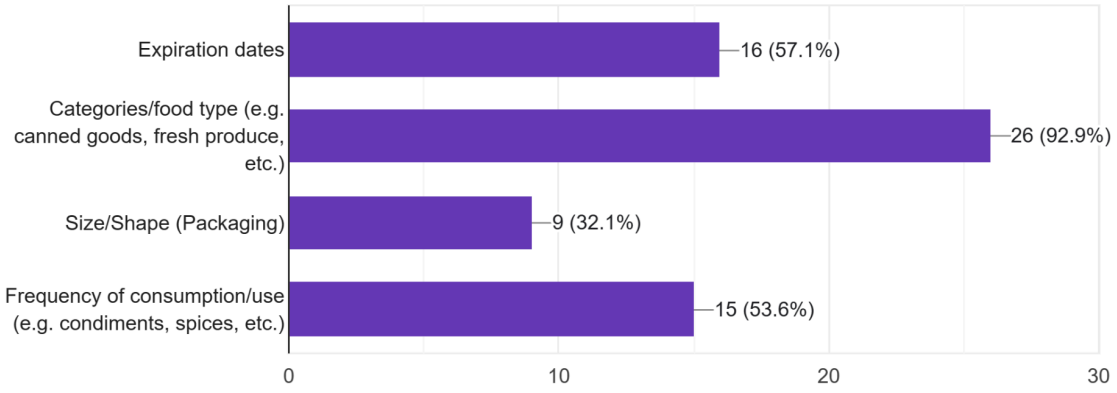


Figure 1.2 *Survey responses when asked "What factors do you consider when organizing your food items? (Select all that apply)"*

According to the responses from the survey, the primary factor (as per Figure 1.2) that 92.9% of participants take into account when organizing their food items at home is their type or category (Hypothesis 1.1). This means that they would organize their food items by grouping similar types of food items which are then stored together in their own storage area away from food items of different types. Secondary factors that the respondents consider are the expiration dates (Hypothesis 1.2) and the frequency of consumption/use. This means that the food items are usually sorted with extra priority given to those with closer expiration dates; also taking into consideration which specific items are used more frequently than others. Finally, the observed least important factor when organizing groceries is the packaging dimensions of the food items. This could be due to most storage areas being able to sufficiently accommodate the different shapes and sizes of packaging that most groceries come in.

These statistics go hand-in-hand with the observed behaviors and actions that were shared in the responses of the interview participants.

Separation of Items by Classification

When asked about their grocery organization at home, almost all interviewees (P1, P2, P3, P4, P7, P8, P9, and P10) exhibited behaviors of storing together food items of the same classification.

P7 mentions that where a grocery item is stored depends on its category, "*...like in terms of if it's snacks, if it's meat, if it's condiments, if it's drinks...*". The general approach their household takes to grocery organization involves refrigerating and freezing perishable food items, while non-perishables are usually stored in the pantry or cabinets.

P2 expresses the same behavior in which they divide the storing of their groceries based on whether the item is wet or dry; where all wet or perishable food are automatically placed inside their refrigerator while dry foods are usually stored inside the cabinets. This method of organization primarily appears to be for both sanitary purposes by actively avoiding cross contamination among the food items, and for better waste prevention by prolonging the shelf life of the food items and avoiding premature spoiling.

Old Stock Before New Stock

Interestingly, another behavior exhibited by a few interviewees, specifically P1, P4, P5, and P10, when asked about how they sort their grocery items, is that they make sure to adjust the organization of the food items at home to force the consumption of old stocks before the new stocks.

For instance, P1 explains that they take out the old stocks from her food storage areas, then place the new stocks at the very back so that the old stocks are the ones up in front, where it is easily accessible for anyone in the household to consume.

P10 also brings up the same behavior when asked about organization, where they called it a "*first in, first out*" basis. This seemed to be done in an effort to prevent the food items they have on-hand from reaching their expiration dates before they are able to be consumed. P4 and P5 mentioned similar behaviors in that they arrange their groceries according to their expiration dates.

Points of Satisfaction and Dissatisfaction

There are some interviewees such as P1 that have expressed satisfaction in organizing groceries, saying that it "*feels good*" when they organize groceries. This generally coincides with the received responses from the survey when asked about whether or not they find organizing groceries satisfying. However, it is important to note that P1 frequently collaborates and communicates with their household's maid with the management of food items, from planning for grocery shopping all the way to leftover management. This could have played a large part in why P1 finds satisfaction in organizing groceries as they always reach a consensus with how to manage groceries without much contradiction.

This point of satisfaction is also something expressed by other interviewees such as P3, where they say that if their personal method of organizing the groceries is what the rest of their family follows, then they feel more motivated to organize the groceries as they feel a sense of importance in the family.

However, several other participants, in turn, have expressed dissatisfaction with organizing groceries, specifically in relation to having to work around leftovers. For instance, P7 states that sometimes if there's too much food already stored in a certain storage area such as the refrigerator, some of the food gets pushed to the back and inevitably forgotten to the point where

it expires before the household members are even able to consume it. The underlying reason for this dissatisfaction seems to be due to a lack of coordination among household members with regards to the management and utilization of leftovers in relation to the new stock grocery items.

Household's Prioritization on Consumption of Food Items

Survey Data Analysis

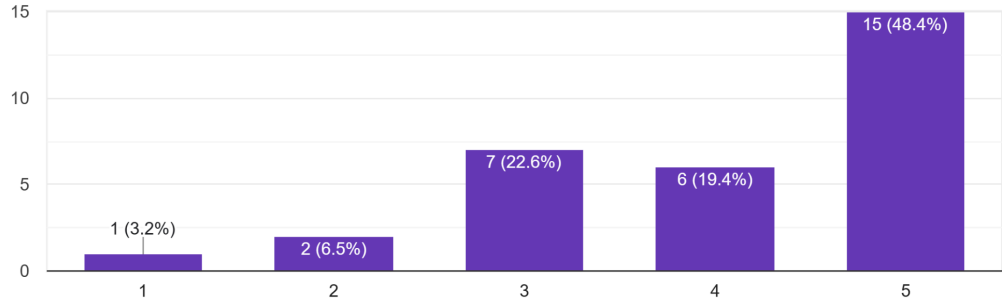


Figure 2.1 *Survey responses when asked "How often do you prioritize consuming food items based on their expiration dates?"*

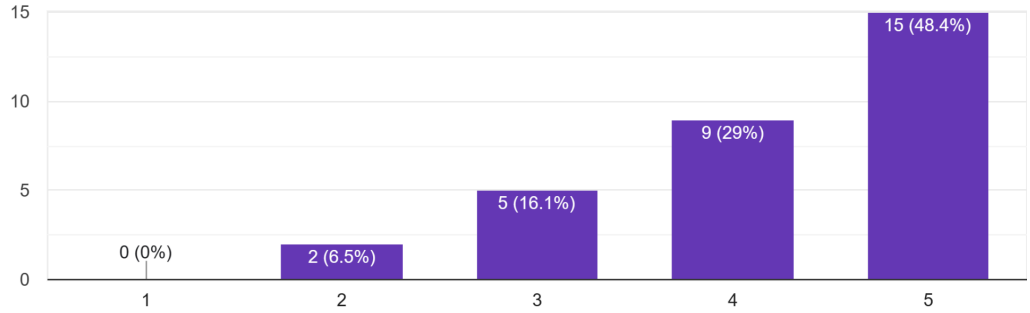


Figure 2.2 *Survey responses when asked "How often do you prioritize consuming food items that are fresh and organic than those that are processed?"*

According to the responses from the survey, the primary factor that majority (53.6%) of the participants take into account when prioritizing which food items to consume first is the expiration date (as per Figure 2.1) (Hypothesis 2.1). This means that they are more inclined to consume food items (such as fresh and organic items, as per Figure 2.2) that expire/spoil faster than those that can last longer unconsumed before they become unsafe to eat.

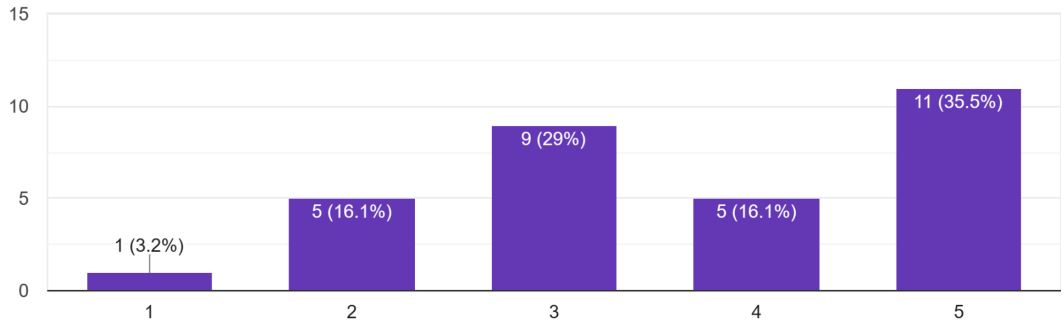


Figure 2.3 *Survey responses when asked "How often do you prioritize consuming food items bought earlier than those bought later?"*

However, it is important to differentiate the idea behind prioritizing food item consumption by expiration date and by date of purchase. When asked about this (as per Figure 2.3), while most of the respondents do prioritize consuming items bought earlier, most of the other respondents seem to be neutral on the matter of food item date of purchase as an important factor. This could be because while it can be assumed that older food stocks are technically closer to their expiration dates, the time gap between the expiration dates of the older food stocks doesn't necessarily differ significantly from that of the newer food stocks.

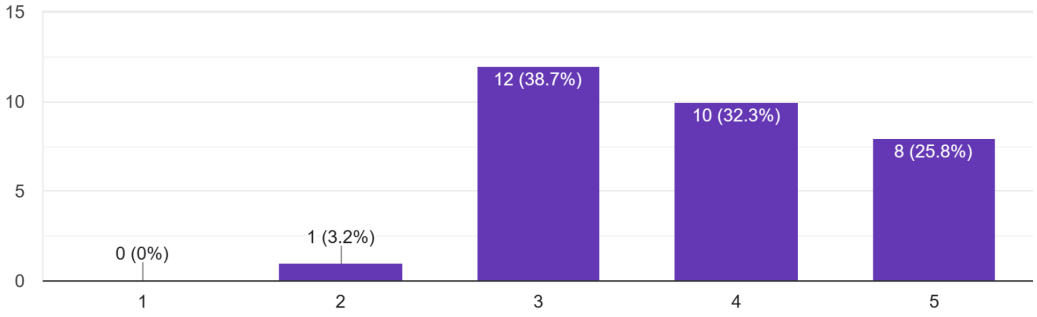


Figure 2.4 *Survey responses when asked "How often do you prioritize consuming food items that you crave at a certain time than those that you do not crave?"*

The secondary factor that respondents consider when prioritizing what to eat is by choosing to eat food that they are craving (as per Figure 2.4) (Hypothesis 2.2), as shown by the data above where there are not a lot of respondents that give the highest priority to cravings, but most still consider it an important factor to take into account when consuming food items.

Cravings and Expiration Dilemma

The general pattern around the responses of the participants when asked about their prioritization on the consumption of their on-hand food items is that they try to find a situational balance between fulfilling cravings and honoring expiration dates, similar to what was observed from the results of the survey.

There were only two observed participants, specifically P2 and P7, who prioritize consuming whatever food item they are craving in the moment. For example, P2's household generally agrees that while it is indeed better to prioritize the consumption of food items based on expiration dates, they tend to prioritize consuming what they are craving for because "*...at the end of the day, if we don't like the food, we technically don't eat the food.*" P7 mentions that their household's method of prioritization of what to consume is primarily based around their cravings, where they explain that "*...if it's not going to expire immediately, then [we prioritize] based on cravings. If there are things that need to be cooked, then maybe you add that.*"

However, it was observed that most of the participants' households (P1, P3, P4, P5, and P8) prioritize their consumption by doing their best to make sure to take the expiration dates of the food items they have at home into account. P8 shares that when it comes to deciding what their household will have for their upcoming meal, they "*just eat whatever is near to their expiration date.*" This behavior is also exhibited by P4, who mentions that their household

prioritizes the consumption of food items based on expiration dates "*...so it won't be a waste of money*."

The responses from the interview participants show a fairly similar pattern to that of the survey respondents, where both sets of participants prefer to prioritize their consumption of food items to be centered around potential expiration/spoilage than what they are craving for.

Waste Prevention via Meal Planning

Another behavior exhibited by some of the participants (P1 and P5) is their proactive measure of planning their meals before cooking food items for consumption.

P1 explains this behavior in their household in strong detail, where they plan the entire household's meals for the entire week. As for the actual ingredients they need to use to prepare their meals, they mention that she takes into consideration food items that are close to expiration and does their best to incorporate those items into the meals, in relation to the "Cravings and Expiration Dilemma" theme.

P5 exhibits a similar behavior where they check their household's food storage areas from time-to-time and notify their fellow household members about food items they see are close approaching their expiration dates. That way, they can prepare meals that incorporate those food items so they can avoid producing unnecessary food waste.

Points of Satisfaction and Dissatisfaction

As expected, multiple participants (P2, P3, P5, P6) have reported being satisfied whenever they consume the food they are craving for. P5 says that in their household, "*it gives us joy when we consume [items] such as soft drinks or chips or ice cream just to elevate our mood*

sometimes." P3 goes even further by saying that they find it rewarding whenever they eat food that was planned to utilize certain ingredients that are close to their expiration dates, where the reward is within the food itself, "*...like good food, good meal at the end of the day.*" It seems that the feeling the participants get when they satisfy their cravings appears to be dependent on whether the rest of the household agrees to follow suit; where they are less likely to feel regret for choosing a food item they were craving for when the rest of their household has made a similar choice.

On the other hand, there are only a few reports of dissatisfaction among the participants (P5 and P10) with regards to prioritizing having to eat leftovers before they can consume freshly-cooked meals. P5 says that they find it discouraging to eat their leftovers because "*...it just feels that you're obliged to eat the food even if you don't like it. Just to avoid the food waste.*" P10 shares a similar thought in that "*...it kind of forces you to eat the exact same thing for like quite a decent amount of time.*" These general sentiments seem to be the case when there is a lack of proper communication among household members not just on what to consume first, but also how much food to prepare for consumption in general.

The Influence of Household Characteristics on Food Consumption Behaviors

Survey Data Analysis

Understanding the different household characteristics and the factors which affect them provides a clearer picture of its influence on food consumption. The researchers conducted a survey to address this question and its corresponding hypotheses.

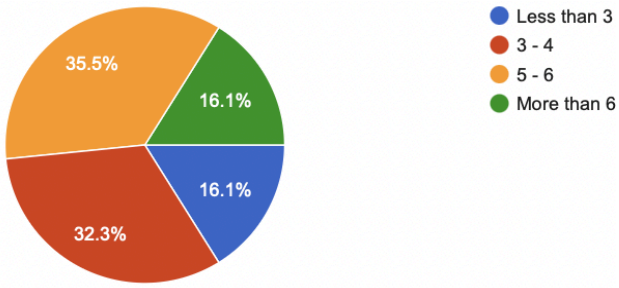


Figure 3.1: *How many people live under the same roof in your household?*

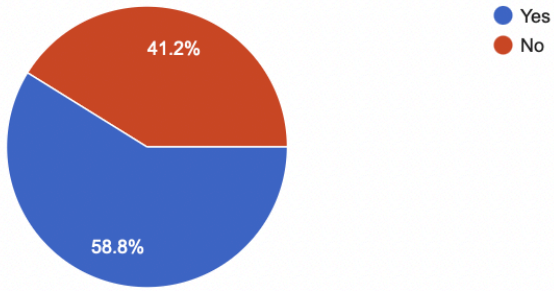


Figure 3.2: *Would you say you tend to buy more food for your children? (Snacks, Juices, etc.)*

The results from the data show that there is a positive relationship between the number of household members and the number of plates per meal (Hypothesis 3.1). On the bracket of less than 3 household members, they average at 2.2 plates per meal, while those in the more than 6 bracket average at 5.13 plates per meal. The data also shows that those with children in their household tend to buy more snacks for their children (Hypothesis 3.2). As such, the data shows that both the number of people in a household and the maturity of people in the household tend to increase the food consumption of the household.

Financial Characteristics

A common theme that's present throughout the participants is the effects of finance on people's experiences, capacity, and enjoyment when buying groceries. For example, P9 with a monthly household income of ₱55,000 indicated that "*it's difficult to buy everything that you want because you have to also consider the prices… Unlike previously, if you have ₱2000, that's already enough. Nowadays, you can't buy much groceries with ₱2000 only.*" Not having the financial capacity to buy whatever they might want or need certainly decreases the groceries that can be bought and in turn decreases the food consumed among these participants. On the other hand, participants like P1 who have a monthly household income of ₱300,000 are not subjected to limiting their food consumption based on their needs and are able to freely consume based on their wants. P1's experiences state that "*Masarap sa feeling yung marami kang stocks.*" Having this kind of financial capacity allows people to buy more and consume more simply because they are able.

Household Characteristics

A household's characteristics also play a role in influencing their food consumption.. For example, participants like P5 who live with 5 other people in the house have to ensure that they are able to provide for everyone and ensure that they all have their basic needs. For them, the number of household members "*really does influence the amount of the food consumed because you have to consider the other members in buying or preparing the food.*" One meal portion per person ends up becoming multiplied by 6, 3 times a day, to account for the needs of the entire household. That is why it becomes easier for participants like P5 to say no to wants and limit their food consumption. P5 stated that "*It is easy to decline right now, especially if you really*

manage your own income. I live alone right now. So last time I did the groceries, it is easy to say no to the wants. So I really do buy stuff considering its usability." This increase in food consumption from an increase of household members rises even further when children are involved. P4 with 9 members in the household said that "*Yes, we do have children in our household. There are three. They often eat a lot and usually have their snacks before having lunch and dinner.*" Snacking for ⅔ of the meals everyday only serves to increase the food consumption in their household.

Satisfaction and Dissatisfaction in the Household

Participants in this study have noted their satisfaction and dissatisfaction with the different situations that they have to cope with due to their household characteristics which lead to different results in their food consumption. Participants like P5 are dissatisfied from having to reject their wants in favor of financial stability. They stated that "*it is also sad that you have to decline, that you have to drop down the things that you want to eat. Just to… Um, save money and then para makaraos ka sa buong month.*" On the other hand, there are others like P3 who are satisfied from being able to save money for their household: "*the satisfaction that it brings when you don't waste the money that you use to buy those foods. Yes. Like, for example, you will think to yourself that the hard work that your parents gave you, like the money that you used to buy those foods, those foods are not wasted.*" In this case, P3 has a larger household income than that of P5 by a difference of ₱30,000.

Household's Prioritization on Food Items When Buying Groceries

Survey Data Analysis

The choice of food items during grocery shopping is a fundamental aspect of household decision-making, influenced by personal preferences, socio-economic factors, and cultural norms. Understanding the characteristics that households prioritize when purchasing groceries is of paramount importance, as it sheds light on the consumption patterns of the household.

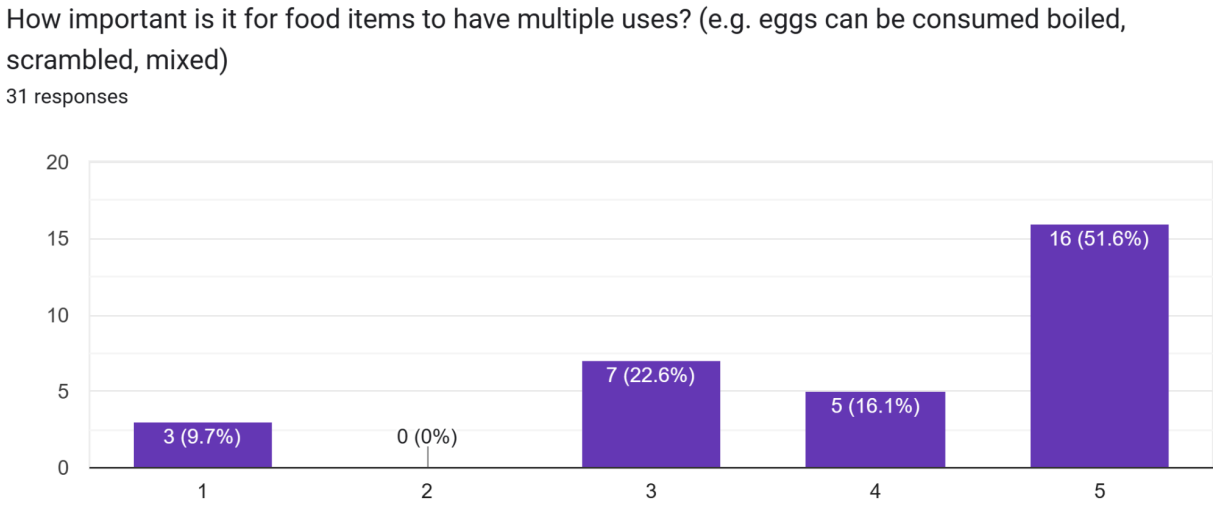


Figure 4.1: *Importance of Food items with multiple uses*

The data from our survey, which examined the importance of food items, reveals insights into consumer behavior. A significant majority of respondents (51.6%) emphasized the importance of versatility in their food choices, indicating a strong desire for items that offer flexibility in their culinary endeavors. This finding aligns with Hypothesis 4.1, which posits that households generally seek versatile food items.

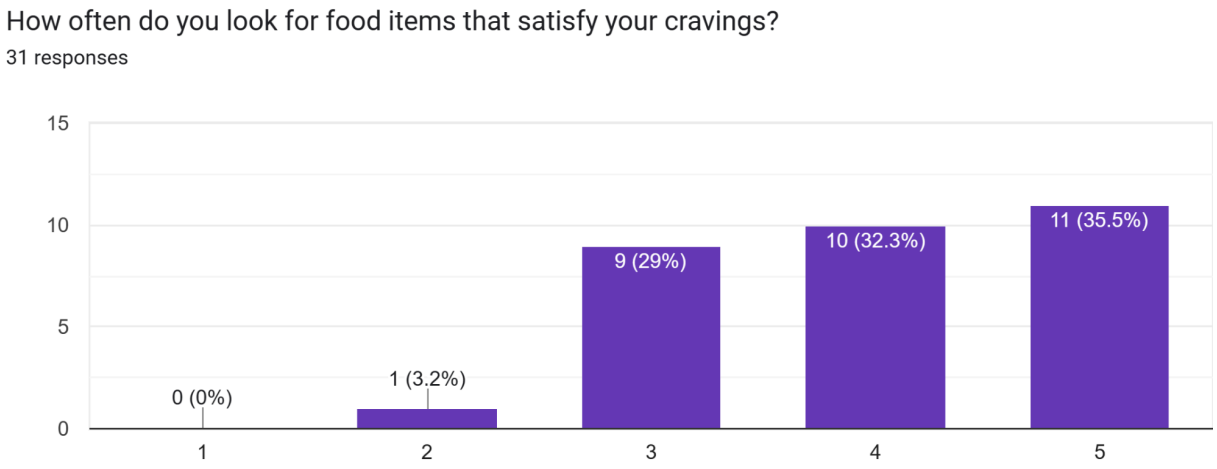


Figure 4.2: *Buying groceries based on cravings*

On the other hand, the frequency of satisfying household members' food cravings during grocery shopping significantly influences consumer behavior. Our survey revealed that 32.3% of respondents frequently seek items to fulfill their cravings, while a majority 35.5% consider it a must. Another 29% do so sporadically, with 3.2% rarely prioritizing cravings. These findings support Hypothesis 4.2, highlighting the importance of craving satisfaction in food choices.

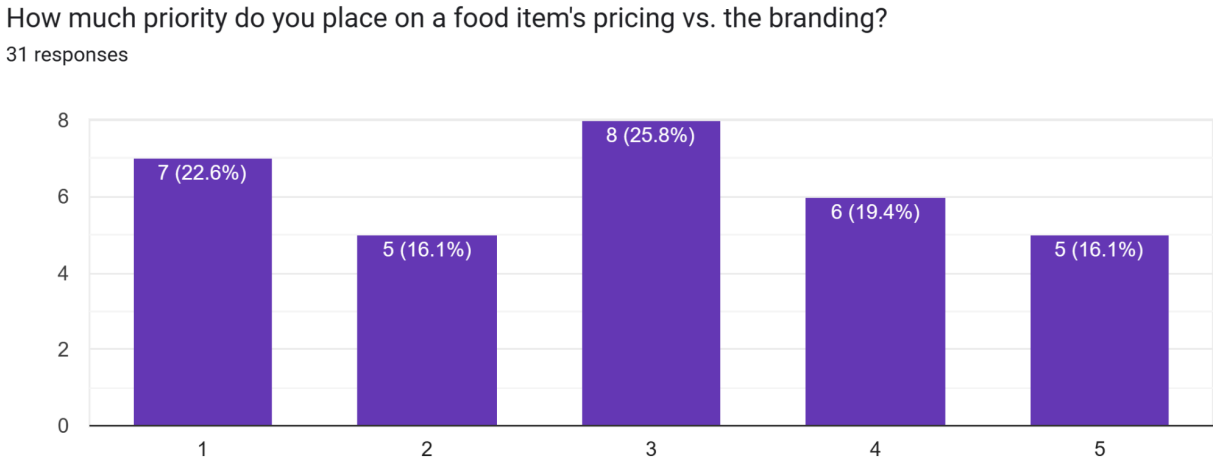


Figure 4.3: *Prioritization of food items based on pricing/branding*

In our survey, we explored the balance between pricing and branding in food selection during grocery shopping. Results indicate that 22.6% prioritize pricing, 16.1% often do, and 25.8% are neutral. Meanwhile, 19.4% often prioritize branding, and 16.1% prioritize branding exclusively. This shows that households make varied choices, striking a balance between cost and branding, aligning with Hypotheses 4.3 and 4.4, which suggest preferences for affordability and branded, high-quality items.

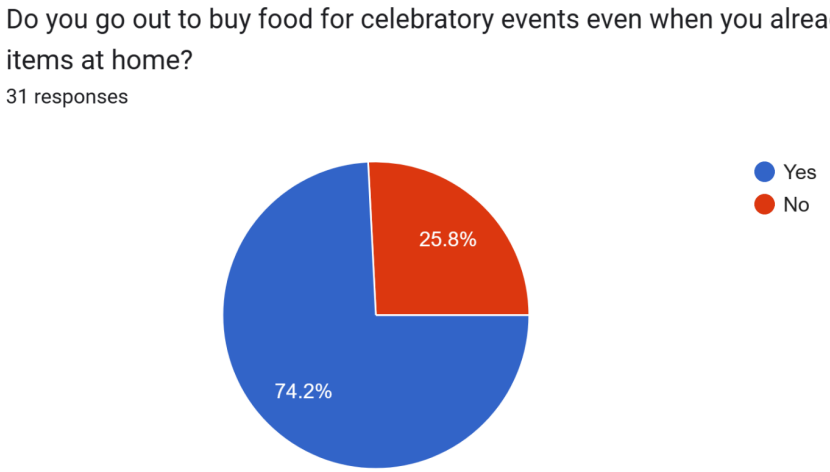


Figure 4.4: *Percentage of Households Purchasing Food Items for Celebratory Events*

Our survey reveals that celebratory events significantly impact household grocery shopping and consumption behavior. A substantial 74.2% of households purchase special food items for such occasions, underlining the influence of events on their buying decisions. In contrast, 25.8% use regular items for special occasions. This reinforces the importance of events in shaping consumption behavior, aligning with Hypothesis 4.5.

Recognizing that households value versatility (Hypothesis 4.1) emphasizes the need for food items that can serve multiple purposes, potentially reducing food waste as items are utilized

more efficiently. Additionally, understanding the role of cravings (Hypothesis 4.2) can guide shoppers in making more informed choices that satisfy their desires without over-purchasing, thus mitigating food waste. Similarly, the awareness of how households consider the trade-off between pricing and branding (Hypotheses 4.3 and 4.4) can inform strategies for cost-effective and sustainable consumption. Finally, understanding the influence of celebratory events in consumption behavior can inform the household of mindful consumption, mitigate over-purchasing, and minimize food waste.

Prioritization of Food Items to Buy

The interviews highlight a general theme wherein households prioritize food items based on various factors, influencing food selection and their direct implications for food waste prevention. Participants' choices are shaped by considerations related to versatility, pricing, branding, cravings, and celebratory events.

When it comes to the aspect of versatility, P8 emphasized its importance, stating that choosing versatile foods means having more options for diverse recipes and culinary endeavors. He stated, "*I buy more based on its usability in different recipes, it would mean that we would have more choices to cook with different ingredients.*" This supports the survey data from which a significant 51.6% of respondents buy food items based on versatility.

Additionally, pricing is taken into consideration and plays a significant role in household food purchases, as indicated by P6. He was observed to buy food items when there were discounts and lower prices, focusing on the economic value rather than brand names. He revealed that, "*If there's a discount on an item, we're most likely to get it, because it costs less anyway.*" However, he also argued that if it's meat instead or if it's a food that most household members like, that's where he prefers the brand. Based on this, there are different scenarios in

which households consider buying food items based on pricing when there are discounts, and preferring branding, if the food item is preferred by the majority in the household. This may also be the reason why the survey data follows a normal distribution with most of the respondents responding with neutral (25.8%) On the other hand, the importance of branding in food choices is expressed by P10. He was observed to buy food items based on brand and past experiences using those brands. He expressed that, "*I shop more for the brands because I know they'll be able to taste better*."

Moreover, the study revealed that individual cravings significantly influence food choices. P7 was observed to buy food items based on the household's cravings and suggestions. She stated, "*There are usually suggestions, like cravings, and there are requests to cook.*" This implies that household members collaborate on buying groceries. P9 also showed this behavior and was observed to have impulsive buying habits, influenced by her cravings. She mentioned "*Sometimes I tend to be an impulsive buyer. So whenever I see items in the grocery, I would buy them.*" This supports the survey data where most of the respondents are inclined to buy food items based on cravings.

The insights from these interviews and their alignment with the survey data emphasize the potential for household collaboration to mitigate food waste. By expressing their preferences, and making efficient choices based on versatility, pricing, branding, and cravings, households can foster sustainable and mindful grocery shopping practices, thus reducing food waste and promoting an overall responsible approach to consumption. Collaborative efforts among household members are promising in addressing the challenge of preventing food waste and promoting more sustainable food practices.

Practices that contributes to food waste in households

Survey Data Analysis

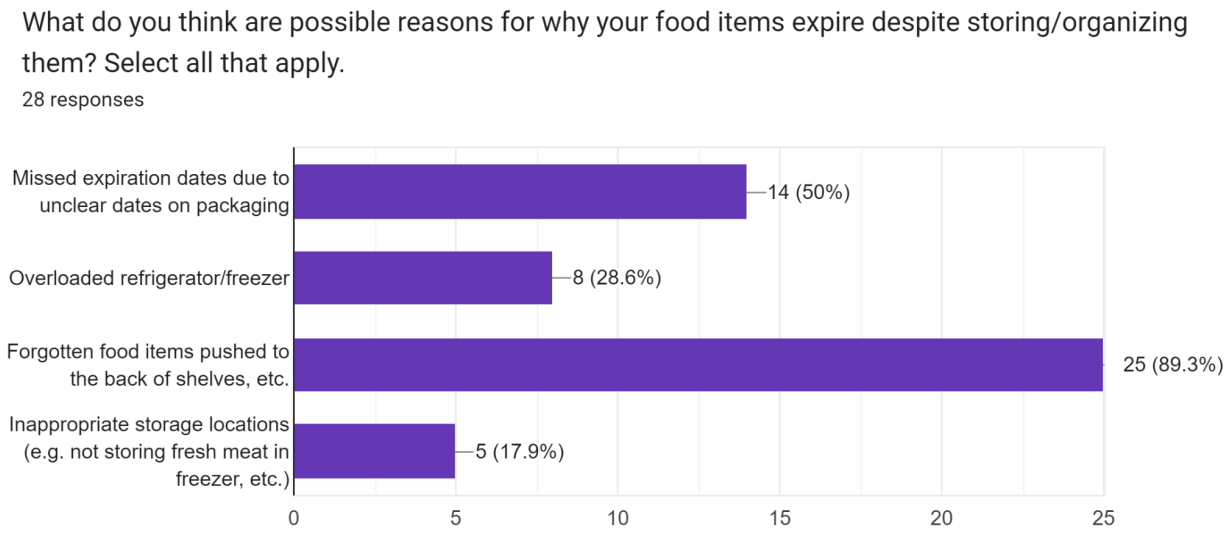


Figure 5.1: *What do you think are possible reasons for why your food items expire despite storing/organizing them? Select all that apply.*

From the survey, the most frequently chosen reasons they thought that caused the expiration of food items were missed expiration dates that were caused by unclear dates on the packaging (50%) and the forgotten food items pushed to the back of their shelves (89.3%). Thus implies that the ones mentioned are the likely major factors that contribute to their household's food waste. Furthermore, missing expiration dates due to being unclear is partly because of the provider and the one looking at the packaging since unclear dates may also be subjective (Hypothesis 5.3). Moreover, forgetting food items due to being pushed back to their shelves also aligns with Hypothesis 5.3 since forgetting is human error in itself and food items being pushed

to the back of their shelves is disorganization, and thus answers Hypothesis 5.3 that disorganization and human error contributes to food waste.

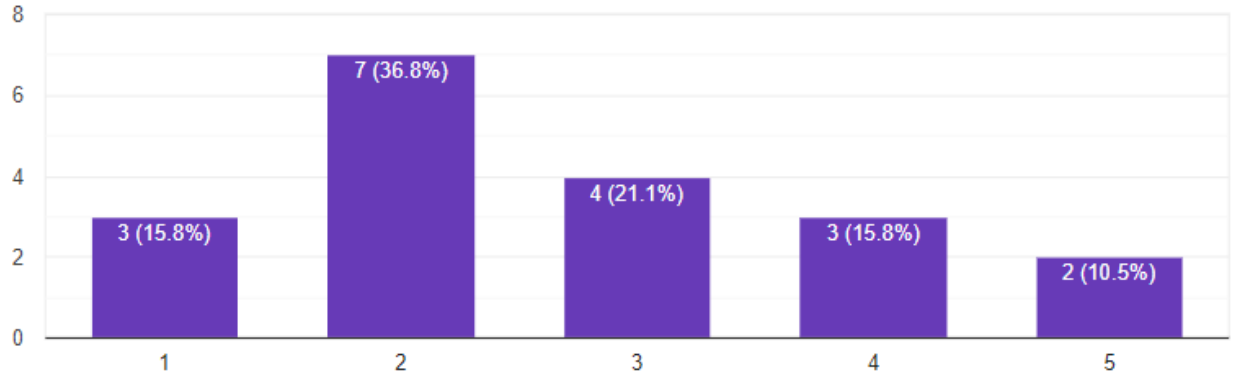


Figure 5.2: *How often do you throw away food items that have expired?*

Results from the survey suggest that more than half of survey responses (specifically 52.6%) expressed that they never, or at the very most, rarely throw away expired food items. To elaborate, the survey question asked if people would throw away expired food items, this should likely be the case for most since expired food items are no longer edible however the survey respondents answered with 52.6% leaning more to never or rarely throwing away expired food items. Thus there may have been a misunderstanding in the way the distribution of the responses were and therefore Hypothesis 5.1 may not have been answered properly.

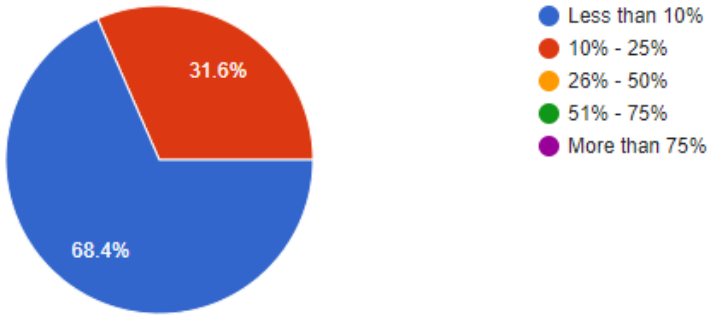


Figure 5.3: *What percentage of your grocery budget do you think is typically wasted due to discarded food?*

The survey data also shows that almost none of the respondents report frequently throwing away food items they no longer like, with 47.4% saying they never do it, 31.6% saying that they seldom do, and the rest saying they sometimes do. This shows that the food waste households produce is seldomly caused by throwing away unwanted food items that haven't expired yet (Hypothesis 5.2).

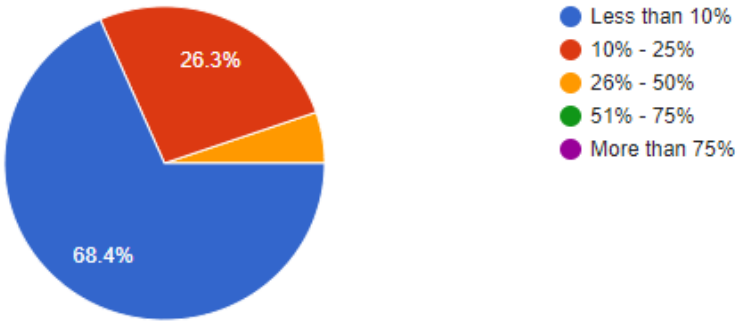


Figure 5.4: *What percentage of your overall household waste consists of food waste (not including natural waste such as bones, fruit peels, etc.)*

The survey respondents report that only about 10% of the entire waste their household produces consists of food waste, not including natural waste such as bone, packaging, and shells. The observed distribution among the respondents is 68.4% of the respondents answered that their food waste is less than 10% of their household waste, 26.3% of the respondents saying it is 10%-25% of their household waste, and the remaining data responding that 26%-50% of their household waste consists of food waste. This suggests that while most households have under 10% of food waste, it is still a cause for concern that 1/4 of the respondents produce more than 10% of food waste.

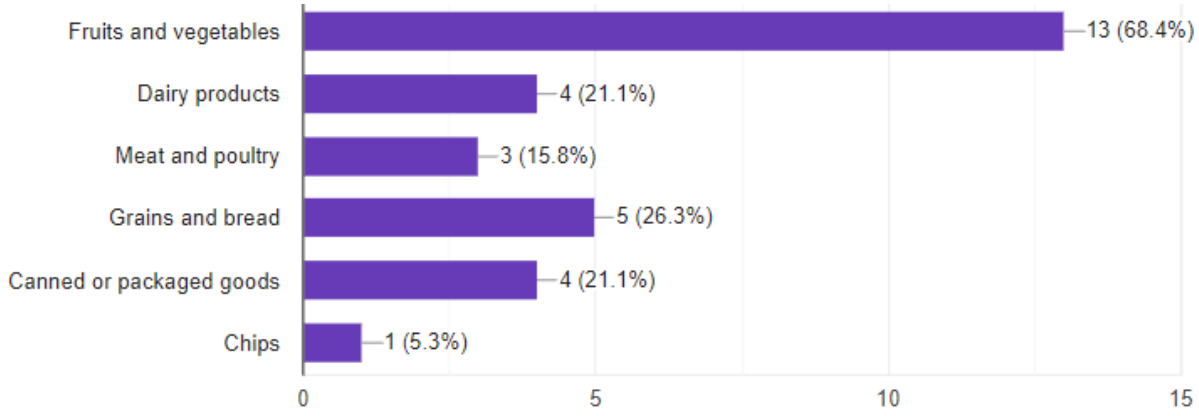


Figure 5.5: *What type of food items are most frequently wasted in your household? Select all that apply.*

When asked about the type of food that is most frequently wasted, 68.4% responded that fruits and vegetables make up most of the food waste, with the other options such as, dairy products, meat/poultry, grains/bread and canned/packaged goods being evenly distributed at 15%-26% each, and chips being the least likely to be wasted. This suggests that fruits and vegetables are the most prone to becoming food waste among households, most likely due to its

relatively fast spoiling and expiration compared to other food types. Thus it'd be probable that when there's food waste, it'd be more likely that it consists of fruits/vegetables.

Unpredictability and Human Restraints

Across the participants, a general theme among them was how their household behaviors and practices affect how much food waste they produce. Many of their responses implied that it is hard to keep track of the food items due to disorganization and human error, which contributes to leftover food and thus may lead to food waste. P7's (similar with P1, P2, P5) circumstance is a good example of this, mentioning that "*Sometimes if there's too much food, some of the food gets pushed to the back. Some food gets forgotten about.*" which implies that food items are forgotten due to being misplaced or having an over-abundance of them which is essentially a product of disorganization and human error. Similarly, P2 (similar with P1 & P3), expresses the same idea that "*if another person used or has used that ingredient, sometimes, they misplace it. So it's hard to keep track where all the things are.*" The consequences of being unable to keep track of food items and forgetting where they were placed would eventually lead to food waste, as stated by P3's (P7 & P9 as well) response "*we're just surprised that we bought something that exists in our cabinet and we see that it's expired*".

Managing household food items with relation to the number household members is also another point of interest, where the interviewees' responses imply its contribution to food waste since many participants like P4 (P1, P2, P3, P5, P7, P8, P9 & P10 have similar sentiments) expressed that it "*influences the amount of food being consumed because of our appetite and depending on the cravings or demand of food that we want to cook.*". P8 also expresses that "*...sometimes it's difficult since some don't eat this kind of food, or they have allergies and stuff.*"

which is similar to P4's response "*...there were a few times that food got thrown out. One of the reasons is that we don't like the food anymore*" in that household members may not eat what they don't like.

Almost all participants, excluding P6, organize their groceries in multiple ways as a way of preventing/mitigating food waste but generally there were two main methods of organization observed from the participants based on the behaviors in the sections "Separation of Items by Classification" and "Old Stock Before New Stock". Another method that almost encapsulates all the participants' sentiments is the management of leftovers, all participants (excluding P9) mention how managing leftovers helps against food waste. Majority of the participants (P1, P2, P3, P4, P5, P6 & P10) give the same idea as that of P2's response "*we can reheat that food for the next day. We usually use the leftover food for our next day lunch or dinner or breakfast.*" and P4's response "*We make sure that there are no leftovers since we share it to our friends and family.*" There were some participants, specifically P6 and P8, that mentioned they do throw away expired leftovers " *...sometimes when we do forget some leftovers, we do tend to throw it out, since it's expired.*", but this is nothing too notable as any expired food item is supposed to be thrown away. The next method that's also frequently observed is related to purchasing groceries and its indirect effect to mitigating food waste. Majority of the participants, specifically P1, P3, P4, P5, P7 & P10 mentioned, mitigate waste prevention through limiting the number of food items they buy. P4 expressed "*We limit the quantity of the ingredients we buy and if it ends up being inadequate, that's where we visit the grocery again.*" which generally means that they'll only go grocery shopping when necessary.

While there are already a number of very prominent methods, there are still few that remain, albeit seem to be less frequently done.. The first approach is meal planning, as was

mentioned by P3, P5, and P10. P10 shares that "*...it's normally already planned by the time we buy them.*", imply that they already plan their meals ahead to avoid not overbuying/disliking the food so as to prevent potential food waste. Another method is the reminding of other household members of impending expirations. P2's (with P5 having a similar statement) says "*we tell people in our household that we need to use this already because it's already gonna get expired*" which implies that reminding household members helps with mitigating food wastage. When it comes to celebratory events that require the cooking of food as to accommodate guests, this method was observed from two participants, P1 and P7, stated that they try lessening food waste by estimating the food to prepare for the guests, such as what was said in P1's response: "*I see to it na alam ko kung ilan muna yung dadating bago kung magluto para hindi masayang ang pagkain.*". Though one respondent, P4, mentioned the idea of reducing the quantity of food items consumed "*As much as possible, we don't tolerate them having leftovers so we teach them to take what they can eat so food won't be wasted.*" which essentially means reducing the amount of food only to what the person can consume.

Referencing the points of satisfaction and dissatisfaction from the themes "Separation of Items by Classification" and "Old Stock Before New Stock", it was observed that while some were satisfied with organizing food items due to having good collaboration with the other members of the household, there were some that were dissatisfied due to potentially not having good communication and collaboration in their households. This affects how motivated people are in organizing their food items as was mentioned before in the behaviors exhibited in the sections "Unpredictability and Human Restraints". Disorganization contributes to food waste thus motivation is a major factor to take into account when examining the overall food waste of households.

Other aspects that participants were observed to be quite satisfied were during the times of consuming food without waste. In P10's response, "*I mean, not rewarding, but I guess more like relief, if you know what I mean, like you're finally done and you could finally have other foods instead of having being stuck eating the same food for for a week in a row.*", it's generally implied (similar to P3 and P7) that they became happy when leftover food was consumed completely.

There were a number of dissatisfied participants with relation to food waste and organization of food items. For food waste, some expressed their concern of throwing out food wastes because it is somewhat saddening for them to not have immediately consumed their food before they became spoiled, as shared by P2 (along with P3 and P9) who says "*need to throw the food which is kind of sad and disheartening*". Some participants (P1 & P4) also expressed their dissatisfaction for the lengthy duration in which food items remain in storage, as observed in P4's response "*It is difficult because sometimes there are instances that the food was way too long kept in the fridge then ending up being spoiled.*" A few participants (P5 & P10) also expressed that they don't want to eat leftover food because "*It's like its already feels that you're obliged to eat this food even if you don't like it. Just to avoid the food waste.*" as shared by P5. One person expressed that it's too tiring to handle leftovers as was stated by P7 "*And sometimes she says that it's tiring.*" which may lead to food waste due to neglect.

From the mentioned information, it can be concluded that the unpredictability and human constraints in managing household food items are primary contributors to food waste. While various strategies like efficient grocery shopping (focusing on only what is needed/planned), proper storage for food items, and leftover management can mitigate waste, their effectiveness and success will heavily rely on effective communication and collaboration among the household

members. Moreover, the emotional responses to food waste further emphasizes the importance of these issues.

FOCUSED DESIGN PROBLEM

A household's production of food waste depends on whether or not they have already existing systems of mitigating food waste. However, despite their systems, numerous factors can still affect the amount of food waste they produce such as their organization methods, prioritization of food item consumptions, the characteristics of their household and the food items they buy. Despite their differences, they exhibit a common behavior of collaborating with other household members on these factors, with the general result behind these collaborations playing a key role in whether or not a household succeeds in avoiding producing excessive food waste. Additionally, household members collaborate with each other in practices against food waste such as meal planning, organization planning, and grocery planning. At some points, they also express satisfaction when they have a successful collaboration and dissatisfaction when they do not. This study aims to contribute a solution to reducing food waste in a household; thus, the researchers will focus on households that have more than one member.

The researchers will focus on the collaboration of the household members and consider **how information should be presented in order to motivate them to collaborate better in managing their household's groceries to avoid the production of food waste.**

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APPENDIX A: Survey Results

Survey Participants 1-5:

	S1	S2	S3	S4	S5
How important is it for food items to have multiple uses? (e.g. eggs can be consumed boiled, scrambled, mixed)	3	5	5	5	1
How often do you look for food items that satisfy your cravings?	4	3	3	5	4
How much priority do you place on a food item's pricing vs. the branding?	1	1	1	2	1
Do you go out to buy food for celebratory events even when you already have existing regular food items at home?	Yes	No	Yes	Yes	Yes
Are there times when food items purchased for celebrations lead to the expiration of regular food items at home?	No	No	No	No	Yes
On a scale of 1 to 5, do you feel that buying	2	3	5	5	5

groceries is more frustrating or rewarding?					
How many people live under the same roof in your household?	Less than 3	5 - 6	More than 6	More than 6	3 - 4
How many plates or servings are there usually every meal in your household?	2	6	6	4 per family	4
Are there any children in your household? (Below 18 y/o)	No	Yes	Yes	Yes	Yes
Would you say you tend to buy more food for your children? (Snacks, Juices, etc.)		Yes	No	No	Yes
What type of food do you buy specifically for your children? (Below 18 y/o) Select all that apply.		Cookies & Biscuits, Juices	Cookies & Biscuits, Juices	Cookies & Biscuits, Juices, Pastry Snacks	Potato Chips, Cookies & Biscuits, Juices, Candy & Chocolate, Pastry Snacks
How often do the children in your household fail to finish their meals due to them eating snacks beforehand?		1	1	2	2
On a scale of 1 to 5, do you feel that taking into account the household's	4	5	4	3	3

needs is more frustrating or rewarding?					
How do you store and organize your groceries (e.g. bread, meat, fruits, etc.)?	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.
What factors do you consider when organizing your food items? Select all that apply.	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)
Do your household members have their own food storage areas?	No	No	Yes	Yes	No
What do you think are possible reasons for why your food items expire despite storing/organizing them? Select all that apply.	Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc.	Forgotten food items pushed to the back of shelves, etc.	Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc., Inappropriate storage locations (e.g. not storing fresh meat in freezer, etc.)	Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc.
On a scale of 1 to 5, do you feel that organizing and the proper storing of	3	5	5	5	5

groceries is more frustrating or rewarding?					
How often do you prioritize consuming food items based on their expiration dates? (i.e. the closer the expiration date, the higher the priority/urgency given)	4	5	4	5	4
How often do you prioritize consuming food items that are fresh and organic than those that are processed? (e.g. fresh items as in meat, produce, and dairy)	2	5	5	5	4
How often do you prioritize consuming food items bought earlier than those bought later?	4	1	4	5	2
How often do you prioritize consuming food items that you crave at a certain time than those that you do not crave?	4	5	3	5	5
How often do you prioritize food items that are for events	2	1	3	5	4

when there is one taking place over regular food items? (For example, choosing pasta and snacks over meat-based meals when there's a party.)					
On a scale of 1 to 5, do you feel that prioritizing what food items to consume is more frustrating or rewarding?	4	5	4	5	3
Age	18-29	30-39	30-39	18-29	18-29
Monthly Household Salary	Below P20,000	P67,001 - P167,000	P34,001 - P67,000	Below P20,000	P67,001 - P167,000
Do you live in a single-family home or in an apartment/condominium unit? Do you rent or do you own the place?	Apartment/Condo unit, renting	Single-family home, owned	Single-family home, owned	Single-family home, owned	Single-family home, owned
What is the main reason your household goes grocery shopping?	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low
How often do you throw away food items that have expired?	4	1	2	5	3
How often do you throw away food items that you no longer want to eat, but are not expired?	2	1	1	3	1
How do you	Store in	Eat them for	Eat them for	Eat them for	Store in

handle leftovers in your household?	refridgerator/freezer for a later date	your next meal	your next meal	your next meal	refridgerator/freezer for a later date
What percentage of your grocery budget do you think is typically wasted due to discarded food?	Less than 10%	Less than 10%	Less than 10%	10% - 25%	10% - 25%
What percentage of your overall household waste consists of food waste (not including natural waste such as bones, fruit peels, etc.)	Less than 10%	Less than 10%	Less than 10%	Less than 10%	10% - 25%
What type of food items are most frequently wasted in your household? Select all that apply.	Fruits and vegetables	Fruits and vegetables	Chips	Canned or packaged goods	Fruits and vegetables
How concerned are you about the environmental impact of food waste?	Somewhat concerned	Very concerned	Somewhat concerned	Very concerned	Neutral

Survey Participants 6-10:

	S6	S7	S8	S9	S10
How important is it for food items to have multiple uses? (e.g. eggs can be consumed boiled, scrambled, mixed)	5	5	5	5	3
How often do you look for food items that satisfy your cravings?	5	5	3	3	3
How much priority do you place on a food item's pricing vs. the branding?	1	3	2	3	3
Do you go out to buy food for celebratory events even when you already have existing regular food items at home?	No	Yes	Yes	Yes	Yes
Are there times when food items purchased for celebrations lead to the expiration of regular food items at home?	No	No	Yes	No	No
On a scale of 1 to 5, do you feel that buying groceries is more frustrating or rewarding?	5	5	3	5	5

How many people live under the same roof in your household?	5 - 6	More than 6	Less than 3	Less than 3	5 - 6
How many plates or servings are there usually every meal in your household?	2	6	2	2	7
Are there any children in your household? (Below 18 y/o)	Yes	Yes	No	No	Yes
Would you say you tend to buy more food for your children? (Snacks, Juices, etc.)	Yes	Yes			Yes
What type of food do you buy specifically for your children? (Below 18 y/o) Select all that apply.	Cookies & Biscuits	Cookies & Biscuits, Juices, Candy & Chocolate, Pastry Snacks			Juices
How often do the children in your household fail to finish their meals due to them eating snacks beforehand?	2	3			2
On a scale of 1 to 5, do you feel that taking into account the household's needs is more frustrating or rewarding?	5	5	5	3	3

How do you store and organize your groceries (e.g. bread, meat, fruits, etc.)?	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.
What factors do you consider when organizing your food items? Select all that apply.	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)
Do your household members have their own food storage areas?	No	No	No	No	Yes
What do you think are possible reasons for why your food items expire despite storing/organizing them? Select all that apply.	Missed expiration dates due to unclear dates on packaging, Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc.	Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Inappropriate storage locations (e.g. not storing fresh meat in freezer, etc.)
On a scale of 1 to 5, do you feel that organizing and the proper storing of groceries is more frustrating or rewarding?	5	4	5	5	5
How often do	5	5	5	3	3

you prioritize consuming food items based on their expiration dates? (i.e. the closer the expiration date, the higher the priority/urgency given)					
How often do you prioritize consuming food items that are fresh and organic than those that are processed? (e.g. fresh items as in meat, produce, and dairy)	5	4	5	5	5
How often do you prioritize consuming food items bought earlier than those bought later?	3	5	5	5	3
How often do you prioritize consuming food items that you crave at a certain time than those that you do not crave?	3	4	3	5	3
How often do you prioritize food items that are for events when there is one taking place over regular food items? (For	3	4	4	4	3

example, choosing pasta and snacks over meat-based meals when there's a party.)					
On a scale of 1 to 5, do you feel that prioritizing what food items to consume is more frustrating or rewarding?	3	5	5	5	5
Age	18-29	30-39	40-49	30-39	30-39
Monthly Household Salary	Below P20,000	P34,001 - P67,000	Over P667,000	P34,001 - P67,000	P34,001 - P67,000
Do you live in a single-family home or in an apartment/condominium unit? Do you rent or do you own the place?	Single-family home, owned	Apartment/Condo unit, owned	Single-family home, owned	Apartment/Condo unit, owned	Single-family home, owned
What is the main reason your household goes grocery shopping?	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low
How often do you throw away food items that have expired?	3	2	2	2	2
How often do you throw away food items that you no longer want to eat, but are not expired?	2	1	2	2	1
How do you handle leftovers in your household?	Store in refridgerator/freezer for a later date	Eat them for your next meal	Store in refridgerator/freezer for a later date	Store in refridgerator/freezer for a later date	Store in refridgerator/freezer for a later date

What percentage of your grocery budget do you think is typically wasted due to discarded food?	10% - 25%	Less than 10%	Less than 10%	10% - 25%	10% - 25%
What percentage of your overall household waste consists of food waste (not including natural waste such as bones, fruit peels, etc.)	10% - 25%	Less than 10%	Less than 10%	Less than 10%	Less than 10%
What type of food items are most frequently wasted in your household? Select all that apply.	Fruits and vegetables	Fruits and vegetables, Dairy products, Meat and poultry, Grains and bread, Canned or packaged goods	Fruits and vegetables, Dairy products	Meat and poultry, Canned or packaged goods	Fruits and vegetables
How concerned are you about the environmental impact of food waste?	Very concerned	Very concerned	Somewhat concerned	Very concerned	Very concerned

Survey Participants 11-15:

	S11	S12	S13	S14	S15
How important is it for food items to have multiple uses? (e.g. eggs can be consumed boiled, scrambled, mixed)	5	1	5	1	5
How often do you look for food items that satisfy your cravings?	5	5	5	4	4
How much priority do you place on a food item's pricing vs. the branding?	1	3	3	3	1
Do you go out to buy food for celebratory events even when you already have existing regular food items at home?	No	Yes	Yes	No	Yes
Are there times when food items purchased for celebrations lead to the expiration of regular food items at home?	No	No	No	No	No
On a scale of 1 to 5, do you feel that buying groceries is more frustrating or rewarding?	3	5	5	5	4

How many people live under the same roof in your household?	3 - 4	3 - 4	5 - 6	3 - 4	5 - 6
How many plates or servings are there usually every meal in your household?	3	2	5	3	1-2
Are there any children in your household? (Below 18 y/o)	Yes	Yes	Yes	No	No
Would you say you tend to buy more food for your children? (Snacks, Juices, etc.)	No	Yes	Yes		
What type of food do you buy specifically for your children? (Below 18 y/o) Select all that apply.	Cookies & Biscuits, Juices	Cookies & Biscuits	Potato Chips, Cookies & Biscuits, Juices, Pastry Snacks		
How often do the children in your household fail to finish their meals due to them eating snacks beforehand?	2	3	2		
On a scale of 1 to 5, do you feel that taking into account the household's needs is more frustrating or rewarding?	3	5	4	3	5

How do you store and organize your groceries (e.g. bread, meat, fruits, etc.)?	I store groceries and organize them.	I store groceries and organize them.	I store groceries but don't organize them.	I store groceries but don't organize them.	I store groceries and organize them.
What factors do you consider when organizing your food items? Select all that apply.	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.)			Categories/food type (e.g. canned goods, fresh produce, etc.), Frequency of consumption/use (e.g. condiments, spices, etc.)
Do your household members have their own food storage areas?	Yes	No			No
What do you think are possible reasons for why your food items expire despite storing/organizing them? Select all that apply.	Forgotten food items pushed to the back of shelves, etc.	Forgotten food items pushed to the back of shelves, etc.			Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc., Inappropriate storage locations (e.g. not storing fresh meat in freezer, etc.)
On a scale of 1 to 5, do you feel that organizing and the proper storing of groceries is more frustrating or rewarding?	5	5			4
How often do you prioritize consuming food items based on	3	5	4	2	5

their expiration dates? (i.e. the closer the expiration date, the higher the priority/urgency given)					
How often do you prioritize consuming food items that are fresh and organic than those that are processed? (e.g. fresh items as in meat, produce, and dairy)	5	4	4	2	5
How often do you prioritize consuming food items bought earlier than those bought later?	3	5	3	2	2
How often do you prioritize consuming food items that you crave at a certain time than those that you do not crave?	3	5	3	4	4
How often do you prioritize food items that are for events when there is one taking place over regular food items? (For example, choosing pasta and snacks over	3	3	3	2	4

meat-based meals when there's a party.)					
On a scale of 1 to 5, do you feel that prioritizing what food items to consume is more frustrating or rewarding?	4	4	4	1	4
Age	50 & above	50 & above	30-39	50 & above	50 & above
Monthly Household Salary	P67,001 - P167,000	P67,001 - P167,000	P67,001 - P167,000	P34,001 - P67,000	P167,001 - P667,000
Do you live in a single-family home or in an apartment/condominium unit? Do you rent or do you own the place?	Single-family home, owned	Single-family home, owned	Apartment/Condo unit, renting	Single-family home, owned	Single-family home, owned
What is the main reason your household goes grocery shopping?	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low	all of the above	Essential food supplies at home are running low
How often do you throw away food items that have expired?	2	1	5	3	4
How often do you throw away food items that you no longer want to eat, but are not expired?	3	3	1	3	2
How do you handle leftovers in your household?	Store in refridgerator/freezer for a later date	Store in refridgerator/freezer for a later date	Eat them for your next meal	Store in refridgerator/freezer for a later date	Eat them for your next meal
What percentage of your grocery	Less than 10%	Less than 10%	Less than 10%	10% - 25%	Less than 10%

budget do you think is typically wasted due to discarded food?					
What percentage of your overall household waste consists of food waste (not including natural waste such as bones, fruit peels, etc.)	10% - 25%	Less than 10%	10% - 25%	Less than 10%	Less than 10%
What type of food items are most frequently wasted in your household? Select all that apply.	Grains and bread	Fruits and vegetables, Grains and bread	Fruits and vegetables, Canned or packaged goods	Meat and poultry, Grains and bread	Fruits and vegetables
How concerned are you about the environmental impact of food waste?	Very concerned	Very concerned	Very concerned	Neutral	Somewhat concerned

Survey Participants 16-20:

	S16	S17	S18	S19	S20
How important is it for food items to have multiple uses? (e.g. eggs can be consumed boiled, scrambled, mixed)	5	4	5	3	5
How often do you look for food items that satisfy your cravings?	4	2	5	4	3
How much priority do you place on a food item's pricing vs. the branding?	2	3	5	5	4
Do you go out to buy food for celebratory events even when you already have existing regular food items at home?	No	Yes	Yes	No	Yes
Are there times when food items purchased for celebrations lead to the expiration of regular food items at home?	No	No	Yes	No	Yes
On a scale of 1 to 5, do you feel that buying groceries is more frustrating or rewarding?	4	3	5	5	2

How many people live under the same roof in your household?	More than 6	Less than 3	Less than 3	3 - 4	5 - 6
How many plates or servings are there usually every meal in your household?	4	3	2	8	6
Are there any children in your household? (Below 18 y/o)	No	No	No	Yes	Yes
Would you say you tend to buy more food for your children? (Snacks, Juices, etc.)				Yes	Yes
What type of food do you buy specifically for your children? (Below 18 y/o) Select all that apply.				Cookies & Biscuits	Cookies & Biscuits, ramen
How often do the children in your household fail to finish their meals due to them eating snacks beforehand?				2	4
On a scale of 1 to 5, do you feel that taking into account the household's needs is more frustrating or rewarding?	3	3	5	4	2

How do you store and organize your groceries (e.g. bread, meat, fruits, etc.)?	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.
What factors do you consider when organizing your food items? Select all that apply.	Categories/food type (e.g. canned goods, fresh produce, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.)
Do your household members have their own food storage areas?	No	No	No	No	No
What do you think are possible reasons for why your food items expire despite storing/organizing them? Select all that apply.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging	Missed expiration dates due to unclear dates on packaging, Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc., Inappropriate storage locations (e.g. not storing fresh meat in freezer, etc.)	Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc.
On a scale of 1 to 5, do you feel that organizing	4	3	5	4	3

and the proper storing of groceries is more frustrating or rewarding?					
How often do you prioritize consuming food items based on their expiration dates? (i.e. the closer the expiration date, the higher the priority/urgency given)	3	3	5	4	1
How often do you prioritize consuming food items that are fresh and organic than those that are processed? (e.g. fresh items as in meat, produce, and dairy)	3	3	5	4	4
How often do you prioritize consuming food items bought earlier than those bought later?	3	3	5	4	3
How often do you prioritize consuming food items that you crave at a certain time than those that you do not crave?	4	3	5	4	2
How often do you prioritize	3	3	5	4	4

food items that are for events when there is one taking place over regular food items? (For example, choosing pasta and snacks over meat-based meals when there's a party.)					
On a scale of 1 to 5, do you feel that prioritizing what food items to consume is more frustrating or rewarding?	3	3	5	5	3
Age	18-29	18-29	30-39	40-49	50 & above
Monthly Household Salary	P34,001 - P67,000	Below P20,000	P34,001 - P67,000	Below P20,000	P20,001 - P34,000
Do you live in a single-family home or in an apartment/condominium unit? Do you rent or do you own the place?	Single-family home, renting	Apartment/Condo unit, owned	Single-family home, owned	Single-family home, owned	Single-family home, owned
What is the main reason your household goes grocery shopping?	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low	Essential food supplies at home are running low	
How often do you throw away food items that have expired?	3	4	2	1	
How often do you throw away food items that you no longer want to eat, but	2	1	1	1	

are not expired?					
How do you handle leftovers in your household?	Eat them for your next meal	Store in refridgerator/freezer for a later date	Eat them for your next meal	Eat them for your next meal	
What percentage of your grocery budget do you think is typically wasted due to discarded food?	Less than 10%	Less than 10%	Less than 10%	Less than 10%	
What percentage of your overall household waste consists of food waste (not including natural waste such as bones, fruit peels, etc.)	26% - 50%	10% - 25%	Less than 10%	Less than 10%	
What type of food items are most frequently wasted in your household? Select all that apply.	Fruits and vegetables	Fruits and vegetables	Dairy products	Fruits and vegetables, Dairy products, Grains and bread	
How concerned are you about the environmental impact of food waste?	Neutral	Very concerned	Somewhat concerned	Very concerned	

Survey Participants 21-25:

	S21	S22	S23	S24	S25
How important is it for food items to have multiple uses? (e.g. eggs can be consumed boiled, scrambled, mixed)	5	3	5	3	4
How often do you look for food items that satisfy your cravings?	5	4	5	4	3
How much priority do you place on a food item's pricing vs. the branding?	2	5	4	4	5
Do you go out to buy food for celebratory events even when you already have existing regular food items at home?	Yes	No	Yes	No	Yes
Are there times when food items purchased for celebrations lead to the expiration of regular food items at home?	Yes	Yes	No	Yes	No
On a scale of 1 to 5, do you feel that buying groceries is more frustrating or rewarding?	4	4	5	5	5

How many people live under the same roof in your household?	More than 6	5 - 6	3 - 4	5 - 6	3 - 4
How many plates or servings are there usually every meal in your household?	5	2	2-3	6	4
Are there any children in your household? (Below 18 y/o)	No	Yes	No	Yes	Yes
Would you say you tend to buy more food for your children? (Snacks, Juices, etc.)		No		No	No
What type of food do you buy specifically for your children? (Below 18 y/o) Select all that apply.		Potato Chips, Juices		Cookies & Biscuits, Pastry Snacks	Milk
How often do the children in your household fail to finish their meals due to them eating snacks beforehand?		1		1	1
On a scale of 1 to 5, do you feel that taking into account the household's needs is more frustrating or rewarding?	4	2	5	3	4

How do you store and organize your groceries (e.g. bread, meat, fruits, etc.)?	I store groceries but don't organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.
What factors do you consider when organizing your food items? Select all that apply.		Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Frequency of consumption/use (e.g. condiments, spices, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Frequency of consumption/use (e.g. condiments, spices, etc.)
Do your household members have their own food storage areas?		No	No	No	Yes
What do you think are possible reasons for why your food items expire despite storing/organizing them? Select all that apply.		Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc.	Forgotten food items pushed to the back of shelves, etc., Inappropriate storage locations (e.g. not storing fresh meat in freezer, etc.)
On a scale of 1 to 5, do you feel that organizing and the proper storing of groceries is more frustrating or rewarding?		2	5	5	5
How often do	3	4	3	5	5

you prioritize consuming food items based on their expiration dates? (i.e. the closer the expiration date, the higher the priority/urgency given)					
How often do you prioritize consuming food items that are fresh and organic than those that are processed? (e.g. fresh items as in meat, produce, and dairy)	3	5	3	5	3
How often do you prioritize consuming food items bought earlier than those bought later?	2	3	3	5	5
How often do you prioritize consuming food items that you crave at a certain time than those that you do not crave?	4	3	3	5	3
How often do you prioritize food items that are for events when there is one taking place over regular food items? (For	4	3	4	5	3

example, choosing pasta and snacks over meat-based meals when there's a party.)					
On a scale of 1 to 5, do you feel that prioritizing what food items to consume is more frustrating or rewarding?	5	2	4	5	3
Age	18-29	18-29	50 & above	40-49	18-29
Monthly Household Salary	P67,001 - P167,000	P67,001 - P167,000	P34,001 - P67,000	P20,001 - P34,000	P34,001 - P67,000
Do you live in a single-family home or in an apartment/condominium unit? Do you rent or do you own the place?	Single-family home, owned	Single-family home, owned	Single-family home, owned	Apartment/Condo unit, owned	Apartment/Condo unit, renting

Survey Participants 26-31:

	S26	S27	S28	S29	S30	S31
How important is it for food items to have multiple uses? (e.g. eggs can be consumed boiled, scrambled, mixed)	4	3	3	4	5	4
How often do you look for food items that satisfy your cravings?	3	4	4	3	5	5
How much priority do you place on a food item's pricing vs. the branding?	4	4	4	3	5	2
Do you go out to buy food for celebratory events even when you already have existing regular food items at home?	Yes	Yes	Yes	Yes	Yes	Yes
Are there times when food items purchased for celebrations lead to the expiration of regular food items at	No	Yes	Yes	No	No	No

home?						
On a scale of 1 to 5, do you feel that buying groceries is more frustrating or rewarding?	4	4	4	5	4	5
How many people live under the same roof in your household?	5 - 6	3 - 4	5 - 6	5 - 6	3 - 4	3 - 4
How many plates or servings are there usually every meal in your household?	6	8	3	4	4	4
Are there any children in your household? (Below 18 y/o)	Yes	No	Yes	No	No	No
Would you say you tend to buy more food for your children? (Snacks, Juices, etc.)	No		Yes			
What type of food do you buy specifically for your children? (Below 18 y/o) Select all that apply.	Cookies & Biscuits		Cookies & Biscuits, Pastry Snacks			
How often do	3		5			

the children in your household fail to finish their meals due to them eating snacks beforehand?						
On a scale of 1 to 5, do you feel that taking into account the household's needs is more frustrating or rewarding?	4	5	4	4	4	4
How do you store and organize your groceries (e.g. bread, meat, fruits, etc.)?	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.	I store groceries and organize them.
What factors do you consider when organizing your food items? Select all that apply.	Categories/food type (e.g. canned goods, fresh produce, etc.), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Frequency of consumption/use (e.g. condiments, spices, etc.)	Categories/food type (e.g. canned goods, fresh produce, etc.)	Expiration dates, Categories/food type (e.g. canned goods, fresh produce, etc.), Size/Shape (Packaging), Frequency of consumption/use (e.g. condiments, spices, etc.)	Expiration dates	Categories/food type (e.g. canned goods, fresh produce, etc.)
Do your household members have their own food storage	Yes	No	No	No	No	No

areas?						
What do you think are possible reasons for why your food items expire despite storing/organizing them? Select all that apply.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc.	Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc.	Overloaded refrigerator/freezer, Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging, Forgotten food items pushed to the back of shelves, etc.	Missed expiration dates due to unclear dates on packaging	Forgotten food items pushed to the back of shelves, etc.
On a scale of 1 to 5, do you feel that organizing and the proper storing of groceries is more frustrating or rewarding?	4	2	2	3	4	5
How often do you prioritize consuming food items based on their expiration dates? (i.e. the closer the expiration date, the higher the priority/urgency given)	5	5	5	5	5	2
How often do you prioritize consuming food items that are fresh and organic than those that are	4	5	4	5	4	5

processed? (e.g. fresh items as in meat, produce, and dairy)						
How often do you prioritize consuming food items bought earlier than those bought later?	2	5	4	4	5	5
How often do you prioritize consuming food items that you crave at a certain time than those that you do not crave?	5	3	4	4	4	3
How often do you prioritize food items that are for events when there is one taking place over regular food items? (For example, choosing pasta and snacks over meat-based meals when there's a party.)	5	3	4	4	4	5
On a scale of 1 to 5, do you feel that prioritizing what food	4	3	3	4	4	5

items to consume is more frustrating or rewarding?						
Age	18-29	40-49	50 & above	50 & above	50 & above	18-29
Monthly Household Salary	P67,001 - P167,000	P167,001 - P667,000	P167,001 - P667,000	P167,001 - P667,000	P67,001 - P167,000	P20,001 - P34,000
Do you live in a single-family home or in an apartment/condominium unit? Do you rent or do you own the place?	Single-family home, owned	Single-family home, owned	Single-family home, owned	Apartment/Condo unit, renting	Single-family home, owned	Single-family home, renting